

PumpRupee

Find the rupees your pumps waste, before you buy a VFD

No flow meter

No shutdown

₹ waste by cause

Team ANS_4X

Schneider Electric Yuva Yodha Energy Tech Hackathon 2026

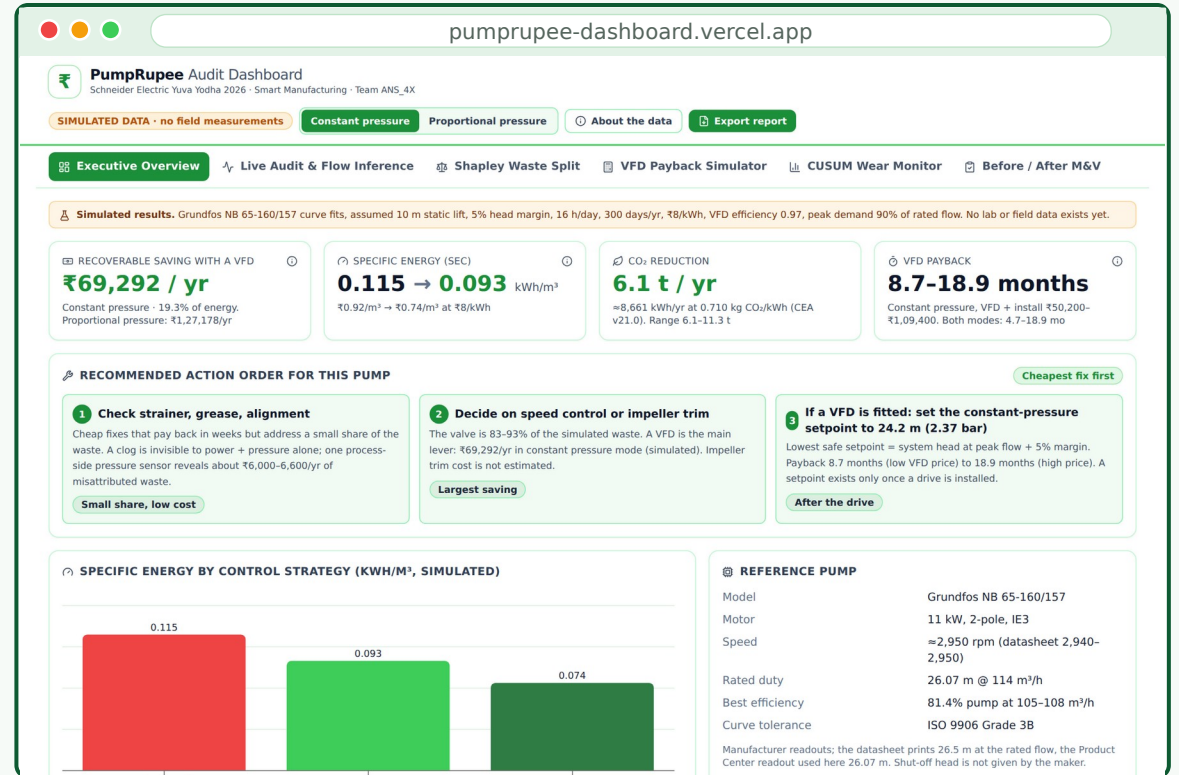
Track: Smart Manufacturing · Electrify · Automate · Digitalise



Live prototype dashboard

<https://pumprupee-dashboard.vercel.app/>

Simulated data · scan the QR code

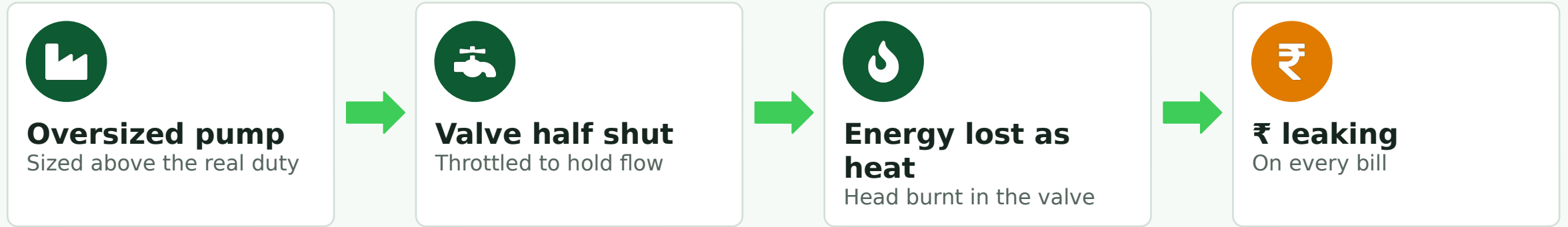


Working prototype dashboard · simulated data

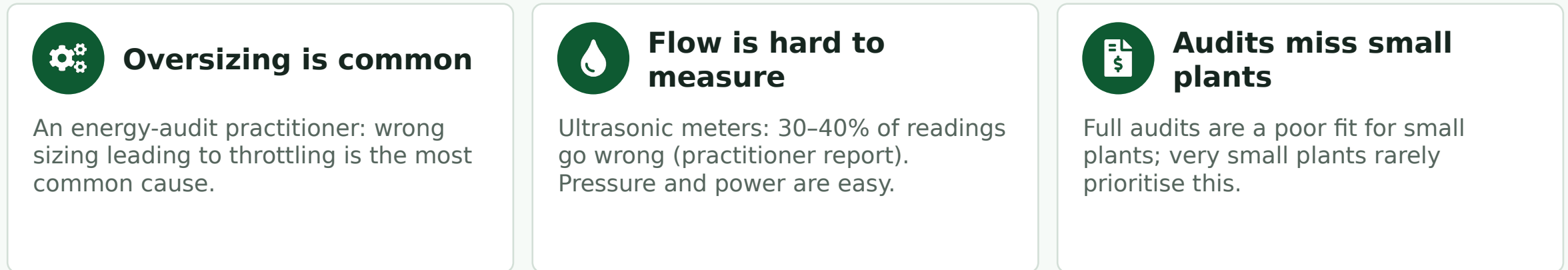
All results are simulated (Grundfos NB 65-160/157, 11 kW). Prices are indicative, not quotes.

Problem Statement

Oversized pumps run throttled. The extra energy is burnt across a valve, every hour, and nobody sees it.



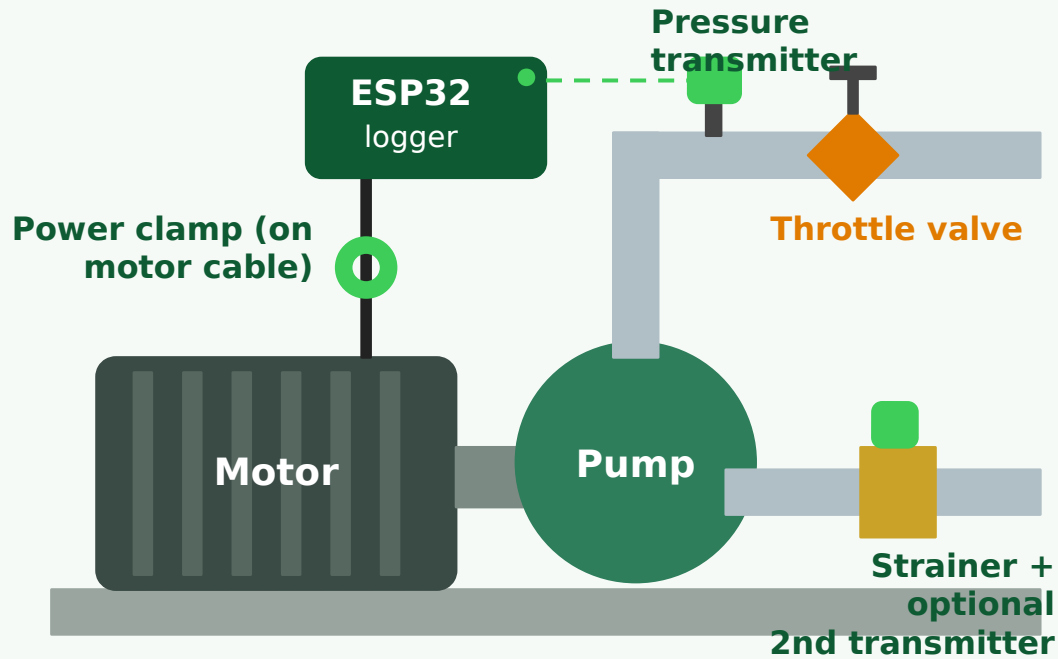
You pay for the pump AND for fighting the valve.



Head (pressure) and electrical power are cheap to measure. Flow is not. PumpRuppee infers flow from the two that are.

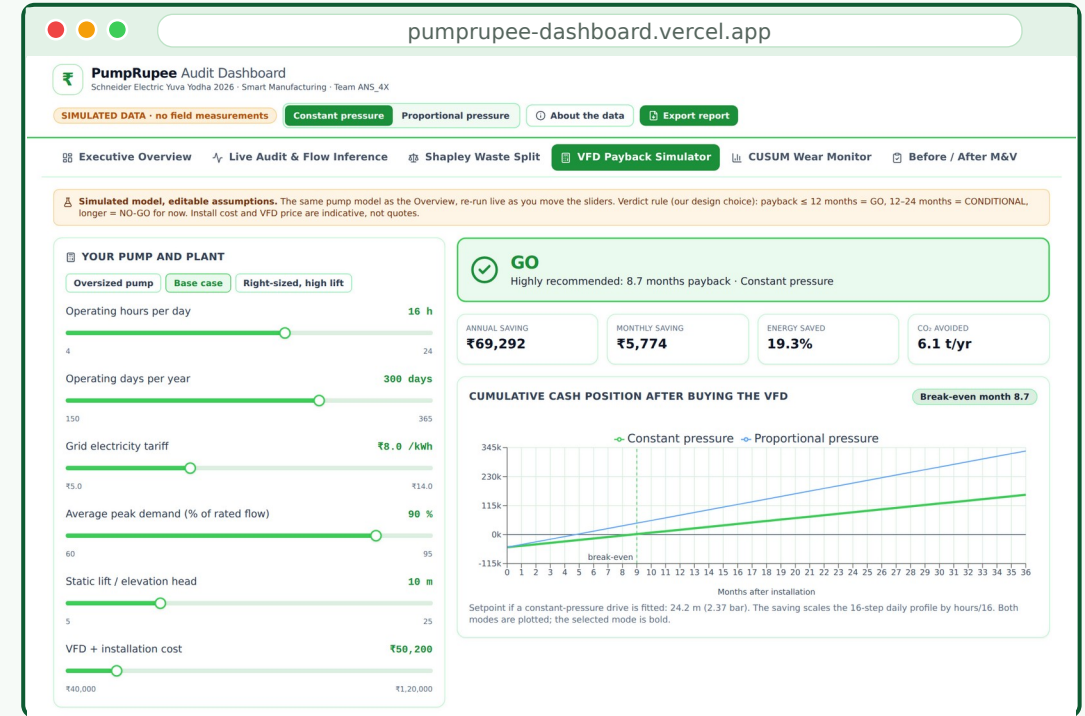
Proposed Solution

A clamp-on kit and a web dashboard turn power + pressure into “₹ wasted per year, and why”.



Concept illustration of the kit on a pump skid

Clamp on → log 7-14 days → ₹ report → the right fix



Working prototype dashboard · simulated data

- ✓ ₹ waste per year, by cause
- ✓ VFD payback verdict for THIS pump
- ✓ Cheapest fix first + safe setpoint
- ✓ Verified saving after the fix

Audits and recommends. Does not control the pump.

Alignment to Smart Manufacturing

Electrify · Automate · Digitalise, at an entry cost an MSME can afford.



Electrify

- Makes motor energy visible with a clamp-on power measurement
- Points owners to the pumps where a right-sized drive pays back



Automate

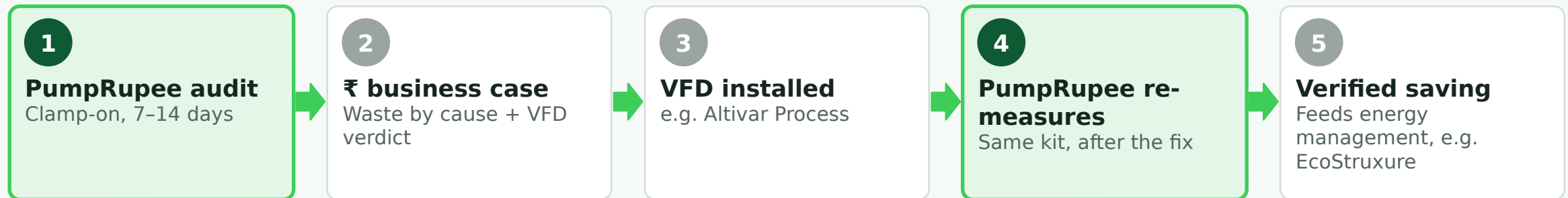
- Automatic diagnosis of waste by cause
- Wear-drift alarm (CUSUM) on the pump
- Lowest safe VFD setpoint; the drive does the controlling



Digitalise

- Logs become a ₹ business case on a dashboard
- Before/after proof of the saving
- Owner keeps the data and can use any drive

How PumpRuppee feeds drives and energy management



A low-cost front door that expands VFD and energy-management adoption into MSMEs that skip audits today. Concept only: no drive or platform integration built.

Key Features & User Journey

Seven steps from clamp-on to a verified saving.

-  **Clamp on** Power clamp on the motor cable
-  **Fit sensors** Pressure on existing gauge ports
-  **Log 7-14 days** ESP32 records power + pressure
-  **Open the dashboard** ₹ report by cause
-  **Apply the fix** Cheapest first; VFD verdict
-  **Re-measure** Verify the saving
-  **Move the kit** Next pump

₹ waste by cause

No flow meter

No shutdown

VFD verdict

Verified savings

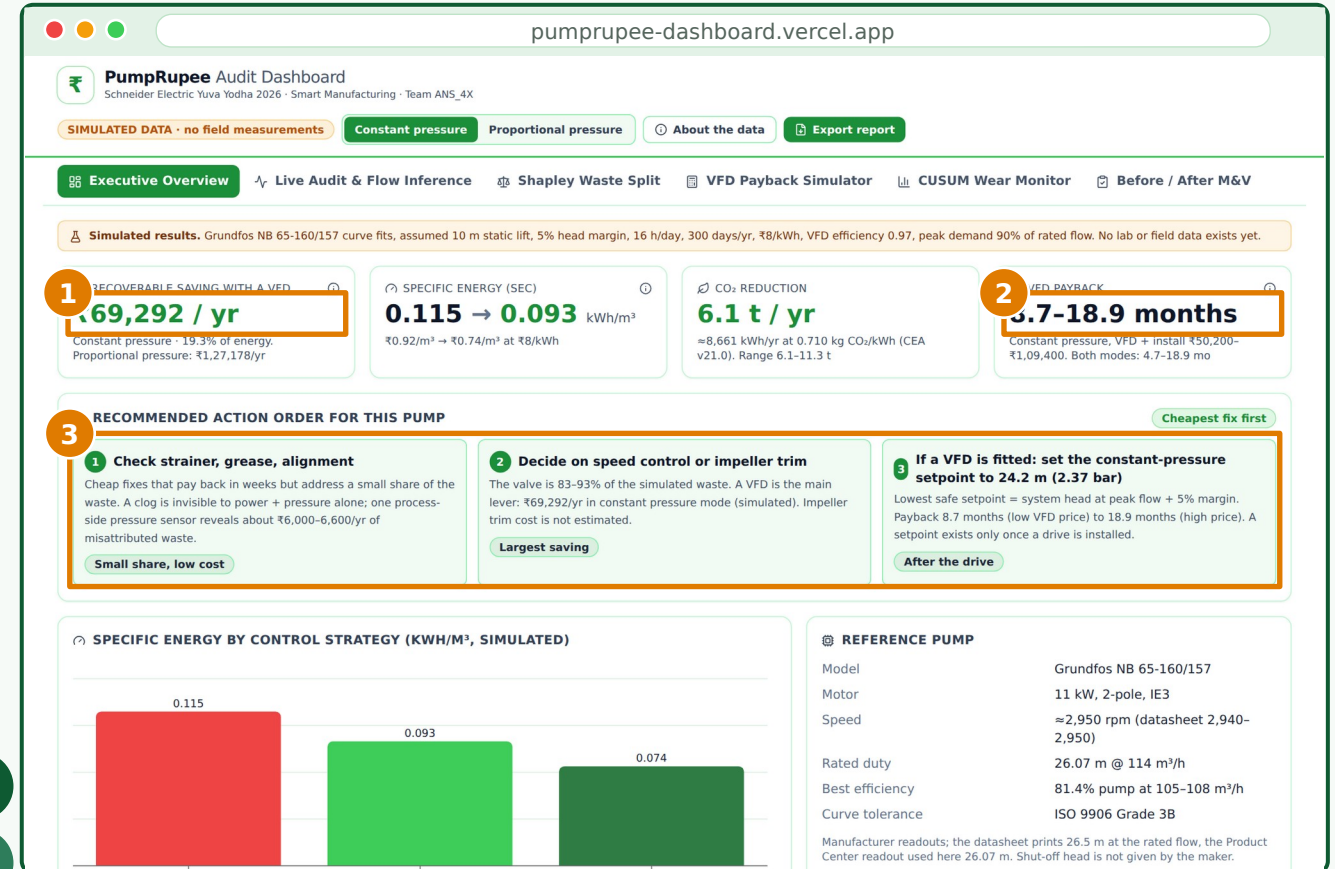
Reusable kit



Live prototype dashboard

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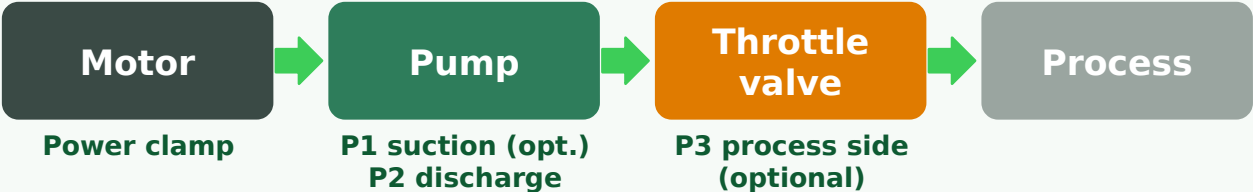
Simulated data · scan the QR code



Working prototype dashboard · simulated data

- ₹ saving per year with a VFD (simulated)
- VFD payback range
- Cheapest fix first, with the safe setpoint

Technical Approach

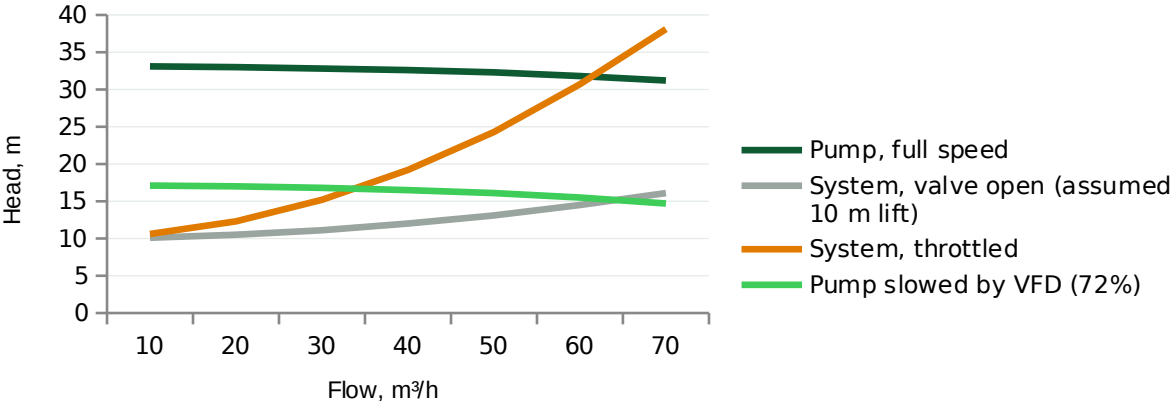


Minimum kit: clamp + P2. Add P3 before a VFD purchase.

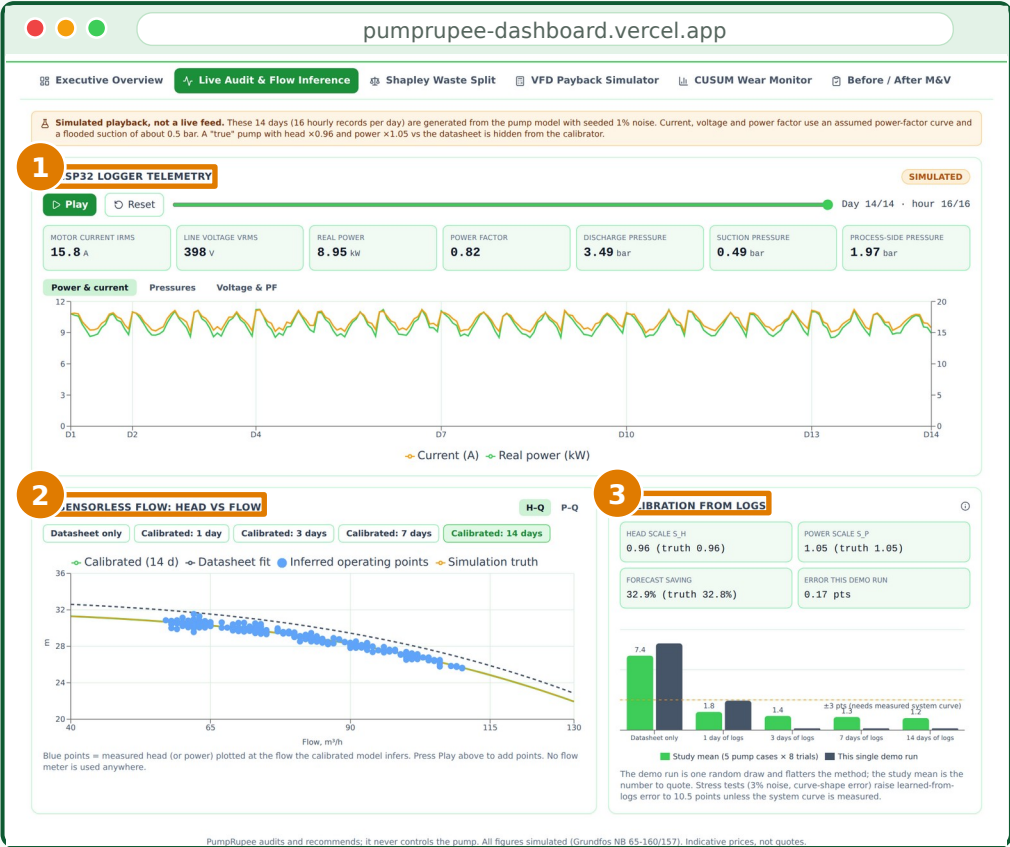


Shapley = fair split of blame between causes. **CUSUM** = early alarm when the pump slowly wears. **SEC** = electricity units per m³ pumped.

Grundfos NB 65-160/157 curves (fit to maker data)



At an off-peak 61.6 m³/h the throttled pump delivers 31.7 m; the pipe needs 15.4 m. The gap is priced in ₹. Simulated; system curve assumed.



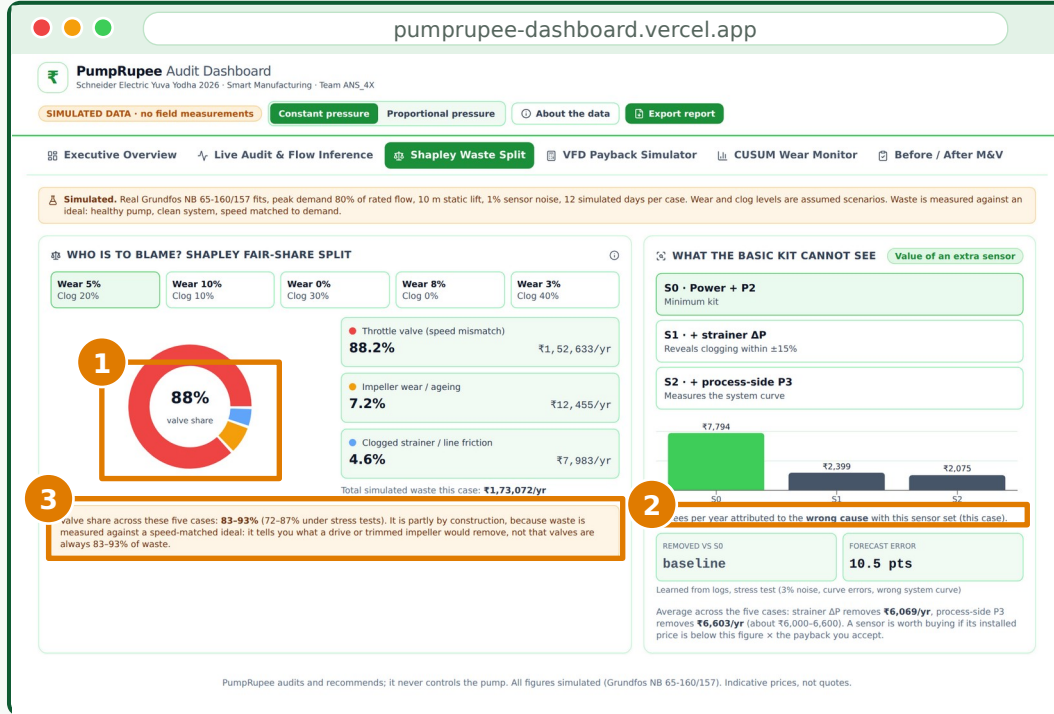
Working prototype dashboard · simulated data

- 1 14 days of ESP32 logs (simulated playback)
- 2 Flow inferred from curve + power + head
- 3 Calibration error vs study mean

About ±3 points only with a measured system curve; otherwise about ±10.

Prototype & Simulated Results

SIMULATED - Grundfos NB 65-160/157, 11 kW · 10 m lift · 5% margin · 16 h/day · 300 days · ₹8/kWh

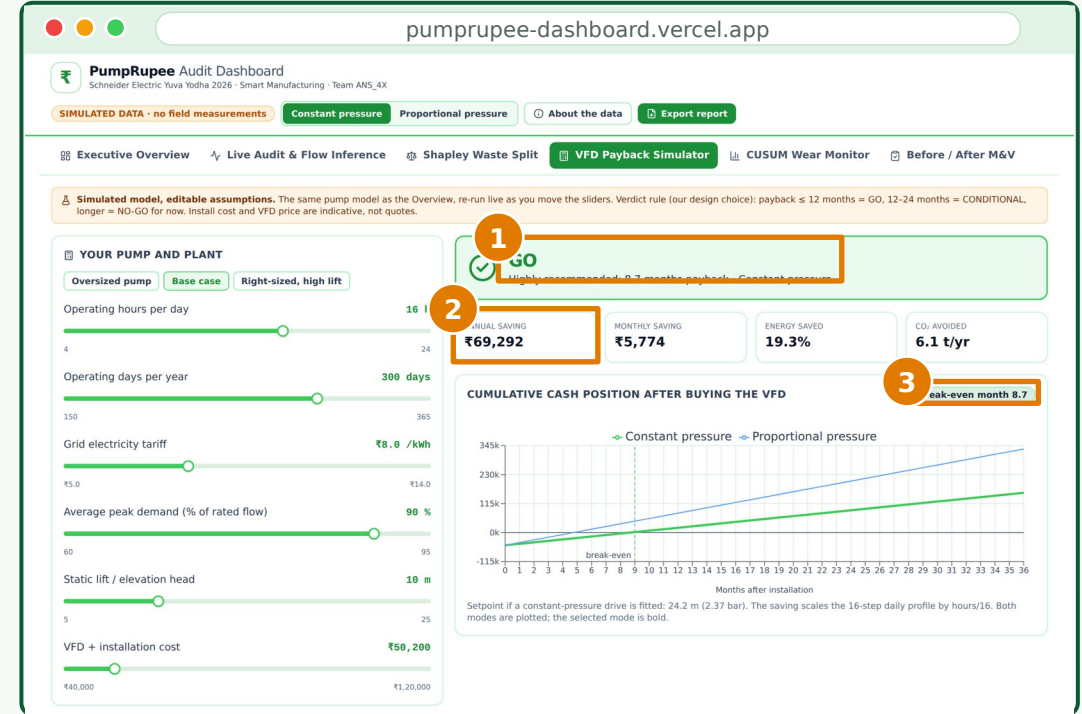


Working prototype dashboard · simulated data

- 1 Valve is the largest share of waste
- 2 ₹/yr attributed to the wrong cause, by sensor tier
- 3 Valve share is partly by construction

₹69,292 / ₹1,27,178
per year (constant / proportional)

83-93%
of waste is the valve



Working prototype dashboard · simulated data

- 1 GO / NO-GO verdict for this pump
- 2 Annual saving in ₹
- 3 Break-even month

4-19 months
VFD payback, oversized pump

±3 / ±10 pts
with / without measured system curve

Innovation

We build on proven tools. The new part is the integration for MSMEs.

Already exists (we build on it)

- KSB PumpMeter, US DOE PSAT
- Samotics (motor-current efficiency)
- Schneider Altivar Process (sensorless flow), EcoStruxure
- Indian IoT vendors and ESCOs
- Shapley split and CUSUM: known methods

Our integration

- ₹ waste split by cause
- Prices the value of an extra sensor
- Low-cost shared kit, no flow meter
- Built-in before/after verification
- Pump-specific VFD verdict before capex

	Flow meter	Manual audit	Full platform	PumpRupree
Upfront cost, small plant	Meter + install	Fee per visit	Higher	₹6,100-13,300
No shutdown	—	✓	—	✓
Answer in ₹	✗	✓	—	✓
Cause breakdown	✗	✓	—	✓
VFD verdict	✗	✓	—	✓
Verified savings	—	—	✓	✓
Fits small MSMEs	—	✗	—	✓

✓ yes ~ partly / varies ✗ no. Our general assessment; PumpRupree's ticks are design targets (simulated).

We found no low-cost shared kit that combines these. Full platforms are the next step after PumpRupree.

Expected Impact

SIMULATED - per 11 kW pump; prices indicative, not quotes

A one-time kit vs a yearly saving (₹)

Kit, one-time and reusable (top tier)

 **₹13,300**

Saving per year, constant pressure

 **₹69,292**

Saving per year, proportional pressure

 **₹1,27,178**

One kit ≈ 5-19% of a year's simulated saving on one pump.



6.1-11.3 t

CO₂ avoided per pump per year (0.710 kg/kWh)



₹50,200-1,09,400

wrong VFD purchase avoided where it won't pay back



4-19 months

payback, oversized pump (2-4+ years if right-sized, high lift)

Who benefits

MSME plant owners

Maintenance teams

Energy auditors / ESCOs

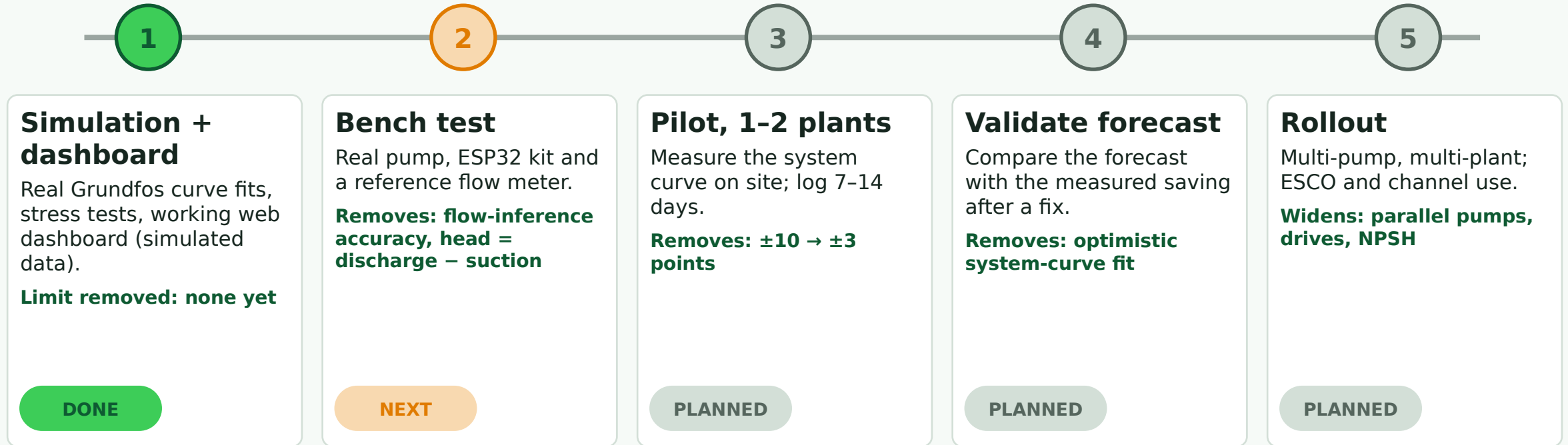
Drive channel partners

India's industrial efficiency goals

Scale-up illustration, not a projection: total = N pumps × per-pump saving. For N = 10: ₹6,92,920-12,71,780 per year and 61-113 t CO₂ per year, using the 11 kW base case. The 54-60% saving for heavily oversized pumps is an upper bound only.

Implementation Roadmap

What is done, what is next, and which limit each phase removes.



Honest limits we validate next

- Simulation and software prototype only: no lab or field data, no real quotes
- About ± 3 points only with a measured system curve; otherwise about ± 10 ; the curve fit is optimistic
- Single pump, steady state. Out of scope: parallel pumps, closed loops, pumps already on drives, min-flow/NPSH, control dynamics

Team Introduction

Team ANS_4X · Schneider Electric Yuva Yodha 2026 · Smart Manufacturing

Insert
photo

Akshaya V G

**Team Lead &
Software/Dashboard**

Data pipeline and the
PumpRuppee web dashboard

Insert
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Adithyaa J

Embedded Hardware

ESP32 logger, power clamp
and pressure-sensor
integration

Insert
photo

Nikilaesh B

Energy Analytics

Pump hydraulics model,
Shapley waste split and VFD
payback logic

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photo

Siva

Subramaniam S
Simulation & Validation

Pump-curve simulation,
CUSUM wear-drift testing, pilot
and roadmap planning



Live prototype dashboard

<https://pumpruppee-dashboard.vercel.app/>

Simulated data · scan the QR code

**Cheap kit. Clever inference.
Answers in rupees.**